Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Software engineer with 8+ years building production data serving infrastructure, ML pipelines, and distributed systems in Java and Python. At Intuit, building ML feature engineering pipelines (scikit-learn, XGBoost) from raw data through low-latency prediction APIs, integrated with downstream AI services. At Entrata (2018--2024), designed and operated high-throughput data serving infrastructure -- eligibility decisioning at scale, ETL pipelines processing 1.6M+ records, and PostgreSQL systems serving 100K+ monthly active users with 99.5% uptime. Strong track record owning architecture, reliability, and operational excellence of production services. AI-native practitioner: LangGraph agentic workflows, Anthropic Agent SDK, daily Claude Code user. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Java / Spring Boot | Python / FastAPI | ML Pipelines & Feature Engineering | Data Serving Infrastructure | PostgreSQL / Advanced SQL | Distributed Systems / Microservices | Async Processing (RabbitMQ) | AWS (Lambda, S3, RDS, SQS) | Kubernetes / Terraform (IaC) | Docker / CI/CD (GitHub Actions) | Observability (Prometheus, OpenTelemetry) | LangGraph / Agent SDK / RAG | scikit-learn / XGBoost | REST APIs / Low-Latency Services

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- ML Pipelines & AI Data Infrastructure -- Virginia, USA

- Build **ML feature engineering pipelines** (scikit-learn, XGBoost) end-to-end: raw data ingestion, feature transformation, model training, and low-latency prediction serving APIs on AWS (Lambda, RDS, SQS).
- Design **data serving infrastructure** for AI/LLM workflows -- input validation, structured output contracts, and golden-dataset regression harnesses ensuring quality and reliability at every release.
- Build agentic AI workflows (LangGraph, Anthropic Agent SDK); instrument services with Prometheus and OpenTelemetry for production observability; deploy via Docker and GitHub Actions CI/CD.

Entrata -- Residential & Affordable Housing SaaS

Sep 2018 -- Jul 2024

Senior Software Engineer -- Data Serving Infrastructure & ML Systems -- Multi-tenant platform, 100K+ MAU

- Designed and owned **high-throughput data serving infrastructure** in Java and Python: eligibility decisioning engine serving 100K+ units at 99.5% uptime -- AMI threshold calculations, income verification, and real-time Approve/Escalate routing with sub-second response requirements.
- Operated **PostgreSQL in production** at scale across multi-schema multi-tenant environments: 200+ query optimizations via execution plan analysis (30% performance improvement across 1M+ records), complex joins, stored procedures, and long-running data model migrations with zero downtime.
- Engineered **ETL and feature transformation pipelines** processing 1.6M+ HUD/Tax Credit records -- federal API ingestion, data normalization, validation, and serving to downstream decisioning systems. Zero data integrity failures over 6 years.
- Built **distributed async data serving** (RabbitMQ) for real-time eligibility and rent updates -- idempotency guarantees, retry logic, and dead-letter handling ensuring consistent low-latency serving across a multi-tenant system at scale.
- Led Docker microservices redesign (50% better scalability); deployed via Terraform IaC on AWS; mentored junior engineers and drove architecture standards across the team. Reviewed 300+ PRs.

PROJECTS

Leasel -- ML Feature Pipeline & Agentic Serving System

Python · LangGraph · RAG · FastAPI · PostgreSQL · Feature Engineering

8-agent LangGraph system with end-to-end feature pipeline: data ingestion, feature extraction and transformation, embedding-based retrieval, confidence scoring, and low-latency serving with human escalation. LLM-as-judge eval CI: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- Multi-Provider ML Inference & Serving Infrastructure

Python · FastAPI · Provider Routing · Low-Latency Serving · CI/CD

15-endpoint FastAPI service with cost-aware routing across five ML/LLM inference backends. Output quality scoring on five metrics; automated CI regression against golden datasets. Addresses ML platform problems: low-latency serving, observability, cost management, and flexible model adoption.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- Clinical ML Serving API

Python · Flask · NLP · REST APIs · RBAC · Docker

Low-latency RESTful serving API for clinical NLP model outputs with fine-grained RBAC and containerized deployment -- regulated-domain ML serving pattern.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 I GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017